Secure Login System (Python Code)

Mini Project / Internship Submission

```python
import hashlib
import json
import os
import getpass

DATABASE_FILE = "users.json"

def load_users():
    if os.path.exists(DATABASE_FILE):
        with open(DATABASE_FILE, "r") as file:
            return json.load(file)
    return {}

def save_users(users):
    with open(DATABASE_FILE, "w") as file:
        json.dump(users, file, indent=4)

def hash_password(password):
    return hashlib.sha256(password.encode()).hexdigest()

def register():
    users = load_users()

    username = input("Enter username: ")

    if username in users:
        print("Username already exists.")
        return

    password = getpass.getpass("Enter password: ")
    confirm_password = getpass.getpass("Confirm password: ")

    if password != confirm_password:
        print("Passwords do not match.")
        return

    users[username] = hash_password(password)
    save_users(users)

    print("Registration successful.")

def login():
    users = load_users()

    username = input("Enter username: ")
    password = getpass.getpass("Enter password: ")

    if username not in users:
        print("Invalid username.")
        return

    hashed_password = hash_password(password)

    if users[username] == hashed_password:
        print("Login successful!")
    else:
        print("Invalid password.")

def main():
    while True:
        print("\n===== Secure Login System =====")
        print("1. Register")
        print("2. Login")
        print("3. Exit")

        choice = input("Choose an option: ")

        if choice == "1":
            register()
        elif choice == "2":
            login()
        elif choice == "3":
            print("Exiting...")
            break
        else:
            print("Invalid choice.")
if __name__ == "__main__":
```

```
main()
```